MIL-E-1/212C
13 August 1971
SUPERSEDING
MIL-E-1/212B
24 March 1965

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, TRANSMITTING

TYPE 3E29

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

Ⓒ DESCRIPTION: Twin-unit beam power amplifier
Outline --- See figure 2
Base --- See figure 2 and Group No. 6 (EIA)
Envelope --- See figure 2
Cathode --- Coated unipotential

Base connections:

Pin No. ---	1	2	3	Ⓒ 4	5	6	7	Top leads	
Element ---	h	2g1	g2	k Beam forming element	hct	1g1	h	1a	2a

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	Ef V (Note 1)	Eb Vdc (Note 2)	Ec1 Vdc (Note 3)	Ec2 Vdc (Note 4)	Ehk v	ec1 v	ib a	ic2 a
Modulator-rectangular	6.3	5,000	-225	850	100	250	10	3.5
Wave modulation (values for both units in parallel)	6.3	5,000	-225	850	100	250	1.5	0.5
TEST CONDITIONS:	6.3	400	Adjust	225	---	---	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter: / Unit:	ic1 a	Pp W	Pi W	Pg2i W	Pg1i W	tp μs	TE °C	Alt ft
Modulator-rectangular	4	15	60	3	1	1.2	175	10,000
Wave modulation (values for both units in parallel)	0.6	15	85	3	1	7.0	160	---
TEST CONDITIONS:	---	---	---	---	---	---	---	---

GENERAL:

Qualification - Required

Ⓒ denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Qualification inspection							
Ⓒ1031	High-frequency vibration (50 Hz)	No voltages; t = 900	---	---	---	---	---	---
	Quality conformance inspection, part 1							
1266	Total grid current	Ec1/Ib = 50 mAdc, t = 30 (see notes 5 and 6)	---	---	Ic1	---	-4.0	µAdc
1256	Electrode current (anode)	Eb = 250 Vdc; Ec2 = 175 Vdc, Ec1 = -11 Vdc (see note 6)	---	---	Ib	38	82	mAdc
1256	Electrode current (screen)	Eb = 250 Vdc, Ec2 = 175 Vdc; Ec1 = -11 Vdc (see note 6)	---	---	Ic2	---	10	mAdc
1261	Electrode voltage (control grid)	Ec/Ib = 200 µAdc	---	---	Ec1	---	-55	Vdc
---	Pulsed operation	Ebb = 5.0 kVdc; Ec1 = -225 Vdc, Ec2 = 850 Vdc; eg1 = +150 Vdc, Rl = 400 ohms (see note 7)	---	---	ib	9.0	---	a
Ⓒ---	Short and discontinuity detection		---	---	---	---	---	---
	Quality conformance inspection, part 2							
1031	Low-frequency vibration (25 Hz)	Eb = 250 Vdc; Ec1/Ib = 10 mAdc; Rl = 2,000 ohms, Ec2 = 225 Vdc (see note 6)	---	---	Ep	---	800	mVac
1036	Bump	Angle = 20°	---	---	---	---	---	---
1331	Direct-interelectrode capacitance	See note 8	---	---	Cg1p	---	0.12	pF
					Cg1, hkg2	12.8	16.2	pF
					Cp, hkg2	5.25	8.75	pF
1301	Heater current		---	---	If	2.0	2.5	A
1336	Heater-cathode leakage	Both filaments energized	---	---	Ihk	---	175	µAdc
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
						MIN	MAX	
	Quality conformance inspection, part 3							
---	Pulse life	Group B; pulsed operation (see note 9)	---	---	---	---	---	---
---	Pulse life-test end point (500 hours)	Pulsed operation test	---	---	ib	7.5	---	a

NOTES:

1. Heaters may be operated in parallel or in series. Maximum variation of Ef - +10 and -5 percent.
2. Instantaneous anode voltage due to transient shall not exceed 5,750 volts. The dc resistance of the supply shall be sufficiently large to limit the short circuit current to 0.5 ampere.
3. Instantaneous grid voltage due to transient shall not exceed -600 volts. The dc resistance of the supply shall be sufficiently large to limit the short circuit current to 0.5 ampere.
4. The dc resistance of the supply shall be sufficiently large to limit the short circuit current to 0.5 ampere.
5. This test to be performed at the conclusion of the holding period.
6. Test each unit separately. Bias unit not under test to -100 Vdc.
7. Use rectangular-wave modulation. Pulse width - 1 microsecond (approximate). Repetition rate - 1,250 pulses per second (minimum). Preheating time - 120 seconds, Ef - 7.0 volts (only). Screen and anode voltages at maximum values to be applied simultaneously. Tap tube during test and reject for prolonged arcs (see figure 1).
8. Test each unit separately. Tie unit not under test to ground. The Cgp shall be measured with a shield 0.75 inch (19.05 mm) in height and 2.475-inch ID (62.87 mm).
9. Forced-air cooling is required.

Custodians:
Army - EL
Navy - EC
Air Force - 80

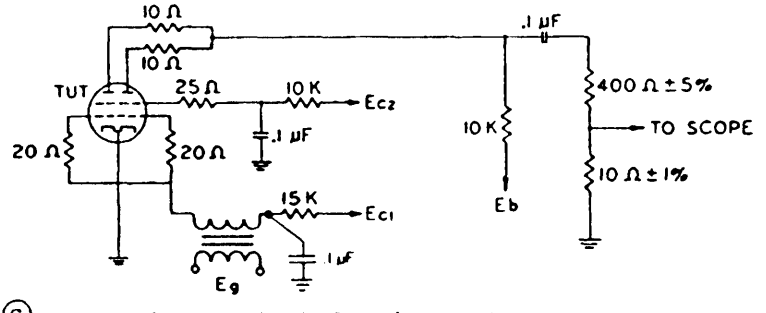
Review activities:
Army - MU
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User activities:
Army - WC
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Preparing activity: Navy - EC

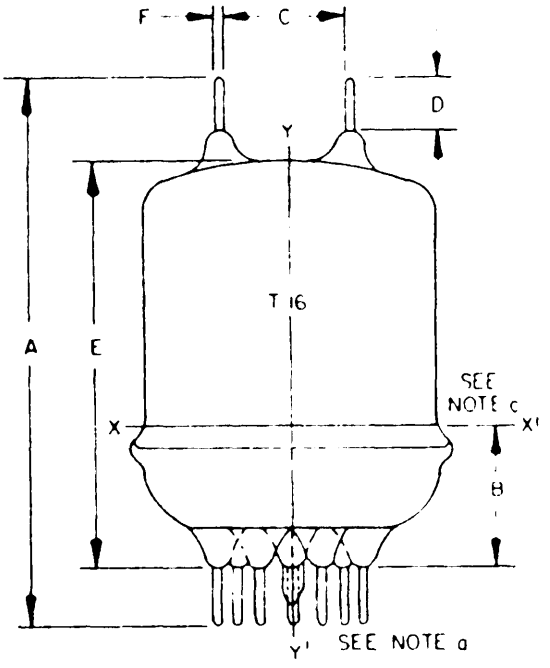
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(Project 5960-2507)



© FIGURE 1. Test circuit for pulse operation.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
A	3.930(99.82)	4.313(109.55)
B	See note b.	
C	See note d.	
D	.375 (9.53)	.625 (15.88)
E	2.875(73.03)	3.250 (82.55)
F	.052 (1.32)	.060 (1.52)

NOTES:

a. The axis YY' is defined as the axis of the base pin gage. (EIA Group No. 6, RS-209A, section 2).

b. Dimension "B" is measured by inserting the tube in the base-pin gage and then lowering a gage plate having a hole 2.063(52.40 mm) + .003(.08 mm) - .000(.00 mm) in diameter until the plate rests on the seal flange at position XX'. The centerline of the hole shall be coincident with the axis YY' within .150(3.81 mm). With the gage plate parallel to the top surface of the base pin gage, the dimension "B" is measured between the bottom surface of the gage plate and the top surface of the base pin gage. This distance shall be .844(21.44 mm) min. and 1.219(30.96 mm) max.

c. Minimum diameter of tube-seal flange will be such that a gage having ID of 2.125 (53.98 mm) + .003(.08 mm) - .000(.00 mm) and thickness of .125(3.18 mm) ± .010 (.25 mm) will not pass the flange when tried at any angle.

d. The anode pins shall be capable of entering a flat plate gage of .375(9.53 mm) min. thickness having 2 holes .120(3.05 mm) ± .0005(.01 mm) in diameter arranged .424 (10.77 mm) ± .001(.03 mm) from a point coincident with the axis YY'. The axes of the holes shall be parallel to YY' and the plane through these axes shall be 90° ± 5' to a plane containing YY' and pin No. 4.

© FIGURE 2. Outline drawing of electron tube type 3E29.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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